

(PostgreSQL License) similar to MIT or BSD, allowing full freedom in commercial and academic use with minimal restrictions. MySQL, on the other hand, uses GPL v2 with a commercial licence from Oracle, which can be problematic for companies that want to embed MySQL without open sourcing their application. MariaDB, also GPL v2 licensed, is governed by the MariaDB Foundation, offering more transparency and community trust than Oracle's stewardship of MySQL. This distinction makes PostgreSQL and MariaDB more attractive for open source and enterprise projects that prioritise licensing freedom.

Usability and ecosystem

Ease of use and learning curve:

When it comes to ease of use and learning curve, MySQL and MariaDB have a clear edge in terms of installation and setup, offering quick and straightforward processes with packages readily available for most operating systems and cloud platforms—a convenience stemming from their shared lineage.

PostgreSQL, while slightly more complex to configure, compensates with production-ready defaults that cater to enterprise-grade deployments. In terms of management tools, MySQL and MariaDB provide excellent support for beginners through user-friendly GUIs like phpMyAdmin and MySQL Workbench, which simplify administration tasks and data handling. PostgreSQL, on the other hand, relies on pgAdmin—a powerful yet occasionally slower tool that includes a query editor, server monitoring, and administrative features.

All three databases are backed by solid documentation and active communities. PostgreSQL is especially noted for its thorough, academically rigorous documentation

that appeals to advanced users, while MySQL and MariaDB provide accessible manuals and guides, with MariaDB emphasising transparency and community-driven development. Across the board, tutorials, forums, and discussions on platforms like Stack Overflow are plentiful, though MySQL generally leads in beginner-friendly content due to its long-standing dominance in web development environments.

Community and market adoption:

In terms of community and market adoption, MySQL remains one of the most widely used databases globally, especially favoured in web applications and content management systems like WordPress, as confirmed by the Stack Overflow Developer Survey (2024).

PostgreSQL, however, has experienced a significant rise in popularity due to its rich feature set and growing support among enterprise-grade applications and modern development frameworks.

Although MariaDB holds a smaller market share, it is highly respected in open source circles and is frequently chosen by developers looking for a MySQL-compatible alternative with more favourable licensing. The developer communities surrounding each of these databases are active and influential—PostgreSQL boasts a technically sophisticated contributor base led by the PostgreSQL Global Development Group and supported by companies like EDB, while MySQL benefits from Oracle's strong backing and a massive user base in LAMP stack deployments. MariaDB thrives under the stewardship of the MariaDB Foundation, appealing especially to users who prefer transparency and independence from corporate control. In terms of platform adoption, PostgreSQL is now a default choice in many cloud services such as AWS RDS, Google

Cloud SQL, and Heroku, reflecting its growing enterprise appeal. MySQL, meanwhile, enjoys broad exposure thanks to its inclusion in most shared hosting services. MariaDB is the default database in several Linux distributions, including Fedora and openSUSE, further cementing its place in the open source ecosystem.

Compatibility and migration:

MariaDB was originally developed as a drop-in replacement for MySQL, allowing most applications and tools built for MySQL to run on it without any modifications. This seamless compatibility is made possible by their shared file formats and client protocols, which significantly reduce migration overhead. However, as both systems have evolved independently, differences in storage engines, optimizers, and feature sets may introduce challenges in more complex environments.

On the other hand, migrating between MySQL/MariaDB and PostgreSQL is considerably more involved due to fundamental differences in SQL dialects—such as how they handle JSON data, window functions, and specific data types—as well as in indexing mechanisms, constraints, and procedural language support. Fortunately, tools like pgLoader, AWS Schema Conversion Tool, and Ora2Pg offer valuable support for schema translation and data migration. Though such migrations are not trivial, they are increasingly pursued by organisations seeking greater ACID compliance, richer SQL capabilities, or enhanced scalability provided by PostgreSQL.

Use case scenarios: What to choose when

The choice between PostgreSQL, MySQL, and MariaDB often hinges on the specific requirements and context of a project.

For web applications, especially content-driven platforms like blogs, forums, content management